

Nirma University

Institute of Technology

Semester End / Supplementary Examination (IR/SPE), February - 2022

Master of Computer Application, Semester - I

3MCA103 Operating System

Roll/
Exam No.

Supervisor's Initial
with date

Time: 2 Hours

Max Marks: 50

Instructions:

1. Attempt all questions.
2. Figures to the right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.

Answer the following Questions

Q.1 What is an operating system? What is 'kernel' in context of an operating system? Differentiate Monolithic kernel and Microkernel and give one OS (operating System) as example in which each type is present. [7]
CO1, BL1

Q.2 Is process and program same? Give your views. Explain all the states and transitions of a **5-state process model** with a neat diagram. [7]
CO2,3 BL2

Q.3 What is meant by **counting semaphore** and **binary semaphore**? Describe bounded buffer producer-consumer problem and write an algorithm for the solution of this problem using counting semaphore. [7]
CO2,3, BL3

Q.4 Explain deadlock in context of operating system? What is the difference among deadlock avoidance, detection, and prevention? Explain **deadlock prevention** in detail. [7]
CO2,3 BL4

Q.5 What is meant by simple paging scheme in context of memory management? [7]
CO2,4, BL6
Consider a simple paging system with the following parameters:
2³² bytes of physical memory; page size of 2¹⁰ bytes; 2¹⁶ pages of logical address space.
a. How many bits are in a logical address?
b. How many bits in the physical address specify the frame?
c. How many entries in the page table?

OR

Q.5 What is dynamic partitioning scheme in context of memory management? Discuss Best fit and Next fit placement strategy with respect to dynamic partitioning scheme taking an appropriate example of your choice. [7]
CO2,4 BL5

Q.6 What is the difference between a **multiprocessor** and a [7]

CO1,3, BL2 **multicore** system? Discuss SMP (Symmetric Multi-processor) architecture in detail using a suitable neat diagram. Also list down potential advantages of SMP over a uniprocessor organization.

Q.7 What is meant by processor scheduling? Explain '**Round-Robin**' and '**Shortest process next**' (SPN) processor scheduling in detail taking an example of your choice. [8]
CO2,3 BL5

OR

Q.7 Discuss **SSTF (Shortest service time first)** and **SCAN** disk scheduling algorithms. [8]
CO2,3 BL6
Also, Consider the disk I/O requests for the tracks received by the disk scheduler as follows:
87, 170, 40, 150, 36, 72, 66, 15.
Initially the disk head is positioned at track 60. Calculate the '**Number of tracks traversed**' in each step and '**Average seek length**' for the SSTF (Shortest service time first) and SCAN algorithms. Draw necessary diagram to show the working of the algorithms.